

Odd-Cycle-Free Latin Squares: Products, Congruence Obstructions, and Constructed Families

27 September 2026

Compact companion; not externally peer reviewed

Abstract

A Latin square is FFF when every two-line permutation in each of its row, column and symbol views has only even cycles. Direct products take the componentwise Boolean OR of the three failure patterns and are group-isotopic exactly when both factors are. Subsquare and congruence obstructions restrict possible constructions. Exact small-order checks exclude orders 6 and 10, while independently checked tables of orders 12 and 14 refute the former power-of-two existence conjecture. Finite-field constructions give additional infinite even-order families. An affine prime-successor family has an exact three-view criterion and parameter classification; a mixed-character-sum estimate gives more than $p/6000$ represented FFF classes at order $p + 1$ for every prime $p \geq 10^{12}$. This does not determine the complete order spectrum.

A row-dependent extension has pattern equal to the OR of its quotient and fibre patterns. When its fibres have no binary three-sorted quotient, all binary quotients descend from the outer square. For an elementary abelian 2-group as outer square, this recovers the layers intrinsically and classifies a fixed family of fibre assignments by affine orbits. A finite audit supplies 514 pairwise distinct FFF main classes of order 20; the construction and Burnside's lemma then give explicit lower bounds in orders $20 \cdot 2^d$, not censuses of those orders. The order-20 quotient freeness uses the computational order-10 exclusion.

An exact position-value coloring formulation supplies a smaller three-view constraint representation with automatic color balance. Proofs, finite computations, and bounded search outcomes are distinguished throughout. Existence at order 18, the complete spectrum, and literature priority for the recent constructions remain unresolved.

Keywords. Latin square; quasigroup; Latin trade; direct product; main class; affine orbit; computational enumeration.

2020 Mathematics Subject Classification. Primary 05B15; Secondary 20N05, 05C70.

Status, attribution, and evidence boundary

This is an AI-generated, internally audited research draft for expert examination, not a submitted or peer-reviewed article. The project initiator reports providing time, a subscription, and local computing resources, but no personal mathematical contribution or capacity to verify the proofs. No accountable scholarly author has yet been established. An empty author field is not an attribution decision, and an AI system is not listed as an author. Rights remain reserved; author responsibility, declarations, literature priority, and external mathematical review are publication gates.

This edition freezes the local evidence available on 27 September 2026 at 10:52 UTC, through register entry C251. Later work in the parallel research session is outside this snapshot. The superseded September 16 PDFs and the September 23 candidate are preserved, not silently replaced.

Theorems with written proofs are distinguished from statements labelled *Computational Theorem*, which require exact finite evidence in addition to a mathematical reduction. Explicitly scoped local exclusions and timeouts are not global theorems. Internal AI-assisted proof review, independent implementations, and machine proof checking are three different forms of checking; none is independent human refereeing. The compact companion imports the order-10 exclusion proved in the full report as an explicit computational dependency. Neither document resolves order 18 or asserts a complete higher-order main-class census.

1 Introduction

An odd cycle in the permutation induced by two lines of a Latin square supports a simple trade: swap the two lines on that cycle. The move preserves the Latin property. In the row and column views it reverses the conventional row–column Latin sign; in the symbol view it preserves that sign and reverses the two parastrophically transported signs. Wanless already gives the underlying three line-parity changes for cycle switches [Wan04, Proposition 1]; the pairwise sign products here are their explicit corollary. This local operation is therefore a natural primitive in parity-based approaches related to the Alon–Tarsi setting, but its three sign conventions must be kept distinct. Bérczi’s experimental program studies such local corrections as ingredients for global constructions [Bér26]; it does not supply the theorems proved here.

We isolate the underlying existence question. For each of the row, column, and symbol views, ask whether some pair of lines induces a permutation with a nontrivial odd cycle. The three answers form a pattern in $\{F, T\}^3$. A square is FFF when all three answers are negative. Equivalently, every union of two link factors in the tripartite representation has component lengths divisible by four. The two-factor/cycle correspondence in one view is classical [WI05, Section 3]; the three-view formulation follows by conjugating the coordinate roles.

The structural results start with two elementary boundaries. Every odd-order Latin square is TTT, whereas the Cayley table of a finite group is FFF exactly when the group is a 2-group. A basis-family closure criterion detects group isotopy. We then show that failure patterns are monotone from Latin subsquares and analyze quasigroup congruences: odd block size forces TTT, while two-element blocks preserve the quotient pattern exactly. Thus a hypothetical FFF quasigroup of order $2p$, for odd prime p , must be congruence-simple in every isotope. This excludes all congruence-imprimitive constructions at order 10 but does not by itself decide the simple case.

The remaining binary line-incidence defect also has an exact subsquare meaning. Nonuniversal vectors in the left binary line kernel are in bijection with half-order subsquares, and the quotient classes group them into complementary four-packs. This yields a closed counting formula and forces maximal binary line rank for every FFF order congruent to 2 modulo 4 and greater than 2. At order 10 this is necessary, not sufficient.

More generally, Moorhouse’s 3-net character formula specializes as follows: a nonuniversal line dependency over a prime field is a balanced cyclic quotient homotopy, and after isotopy its fibres form a quasigroup congruence. Consequently every Smith torsion prime divides the order. For order $2p$, with p odd prime, the congruence obstruction shows that

even one F view forces a saturated integer line-incidence lattice.

The group construction is not the whole phenomenon. A twisted binary extension of C_4 gives an explicit non-group-isotopic FFF loop of order 8. Moreover, direct products satisfy the exact componentwise formula

$$\text{pat}(L \times K) = \text{pat}(L) \text{ OR } \text{pat}(K),$$

and a product is group-isotopic exactly when both factors are. These theorems turn one order-8 example into infinite power-of-two families.

The order-8 computation scans the cited main-class census. McKay, Meynert, and Myrvold enumerated the main classes of small Latin squares [MMM07]; the associated ANU file contains one representative of each of the 283657 order-8 main classes [McK]. The scan finds exactly 230 FFF main classes. Only 5 are group-isotopic and 225 are not, and all eight ordered failure patterns occur. Independently, fresh exact-cover generation of all 9408 reduced order-6 squares finds no FFF square and matches the official ANU set exactly.

Combining theory and computation proves that every order 2^m , $m \geq 3$, realizes all eight patterns and contains a non-group-isotopic FFF square. We do not claim unrestricted direct-product cancellation on isotopy or main classes. Section 10 establishes more specific distinction and classification results under explicit hypotheses. The accompanying complete research report proves by exhaustive master searches that order 10 has no FFF square; its proof is not repeated in this compact companion, but its result is explicitly used in the order-20 quotient-freeness argument and in $h(5) = 2$. A reduced order-12 FFF witness, independently checked in all three views, refutes the former power-of-two conjecture; orders 14 and 36 have further positive witnesses. The classical prime round-robin table is FFF exactly for $p \equiv 3 \pmod{8}$, supplying infinitely many non-power-of-two orders; with the separately audited order-10 exclusion, its order-20 instance gives $h(5) = 2$. A written three-point lift proof gives an FFF square of order $m2^{\text{ord}_m(2)}$ for every odd $m > 1$, without determining the least dyadic exponent in general. Order 18 remains the next undecided small even non-power-of-two order.

The row-fibred extension in Section 10 generalizes the product pattern formula. Quotient-free fibres make the quotient layers intrinsic; over an elementary abelian outer group this gives an exact affine-orbit classification within a fixed family. Exact finite controls supply the 514 order-20 input classes. The resulting large counts come from a construction theorem and Burnside's lemma, not direct scans of every output square.

The longer companion develops additional cycle-parity, subset-action, primitive-group, and flag-surface machinery, gives the order-10 completeness argument, and records the restricted order-18 search boundary. The general product, row-fibred, and quotient-descent proofs do not depend on that machinery; their stated order-20 specialization does depend on the order-10 computation.

2 Definitions and invariance

Let X be a set of size n . A *Latin square* on X is a map $L: X \times X \rightarrow X$ such that every row and every column is a bijection. For $a, c, u \in X$, define the row, column, and symbol-line bijections

$$r_a(c) = L(a, c), \quad q_c(a) = L(a, c), \quad s_u(c) = \text{the unique } a \text{ such that } L(a, c) = u.$$

For two lines in the corresponding view, put

$$\rho_{a,b} = r_b^{-1}r_a, \quad \kappa_{c,d} = q_d^{-1}q_c, \quad \sigma_{u,v} = s_v^{-1}s_u.$$

All three permutations act on an n -element complementary coordinate set.

Lemma 2.1. *If the two lines are distinct, the corresponding induced permutation is fixed-point-free.*

Proof. A fixed point of $\rho_{a,b}$ would give $L(a, c) = L(b, c)$ in one column, contrary to the Latin property. The column case is identical. A fixed point of $\sigma_{u,v}$ would say that the same cell contains the two distinct symbols u and v . $\square$

Definition 2.2. A view of L *fails* if one of its induced two-line permutations has an odd cycle of length greater than one. Write

$$\text{pat}(L) = (\text{row}(L), \text{col}(L), \text{sym}(L)) \in \{F, T\}^3,$$

where T means that the view fails. A square is *FFF* if $\text{pat}(L) = \text{FFF}$.

The terminology “fails” is chosen from the perspective of the local-trade program: a T view supplies an odd-cycle trade, while an F view does not.

Definition 2.3. Fix two rows $a \neq b$ and a union C of cycles of $\rho_{a,b}$. The associated *two-line trade* swaps the entries in rows a, b in precisely the columns in C . Column- and symbol-view trades are defined by permuting the coordinate roles.

Proposition 2.4. *A two-line trade produces another Latin square. Define the three line-sign products*

$$R(L) = \prod_a \text{sgn}(r_a), \quad C(L) = \prod_c \text{sgn}(q_c), \quad S(L) = \prod_u \text{sgn}(s_u),$$

and put

$$\epsilon_{RC} = RC, \quad \epsilon_{RS} = RS, \quad \epsilon_{CS} = CS.$$

The first is the conventional row-column Latin sign, while the other two are its parastrophic companions. If a trade has support size k and $\delta = (-1)^k$, then its sign ratios are

trade view	R'/R	C'/C	S'/S	$\epsilon'_{RC}/\epsilon_{RC}$	$\epsilon'_{RS}/\epsilon_{RS}$	$\epsilon'_{CS}/\epsilon_{CS}$
row	1	δ	δ	δ	δ	1
col	δ	1	δ	δ	1	δ
sym	δ	δ	1	1	δ	δ

Consequently an odd row- or column-view trade reverses the conventional ϵ_{RC} , whereas an odd symbol-view trade preserves ϵ_{RC} and reverses ϵ_{RS} and ϵ_{CS} .

Proof. Write $\pi = \rho_{a,b}$. Since $r_a(c) = r_b(\pi(c))$ and C is π -invariant, the two rows contain the same symbol set on C . Swapping those entries therefore preserves both row symbol sets, while each affected column merely swaps two entries. Thus the result is Latin.

Extend $\pi|_C$ by the identity outside C . The two changed row permutations are obtained by precomposition with π and π^{-1} , so their sign changes cancel and $R'/R = 1$. Each of the $|C|$ affected column permutations is composed with a transposition, so $C'/C = (-1)^{|C|}$. The same support contains the same $|C|$ symbols in the two rows; each associated symbol-position permutation swaps those two rows, giving $S'/S = (-1)^{|C|}$. This proves the first table row. Exchanging rows and columns proves the second.

For a symbol-view trade, each support column swaps the two selected symbols, and the cycle relation pairs the affected cells in exactly the same number of rows. Hence $R'/R = C'/C = (-1)^{|C|}$. The two changed symbol-position permutations differ from their predecessors by mutually inverse permutations on the support, so their sign changes cancel and $S'/S = 1$. Multiplying the appropriate line-sign ratios proves the last three columns. $\square$

Proposition 2.5 (Tripartite link formulation). *Regard the cells of L as the triangle decomposition*

$$\{\{a, c, L(a, c)\} : a, c \in X\}$$

of the complete tripartite graph on the row, column, and symbol copies of X . The links of the vertices in each part form a 1-factorization between the other two parts. An induced two-line cycle of length ℓ corresponds bijectively to an alternating cycle of length 2ℓ in the union of the two associated link factors. Hence L is FFF if and only if every two-factor union in all three link factorizations has only cycle lengths divisible by four.

Furthermore, the total number of 4-cycles among the two-factor unions is the same in all three views and equals the number of intercalates of L .

Proof. For a row a , its incident triangles give the perfect matching $M_a = \{\{c, r_a(c)\} : c \in X\}$ between columns and symbols. For distinct rows a, b , alternately following an edge of M_a and an edge of M_b sends a column c to $r_b^{-1}r_a(c) = \rho_{a,b}(c)$. Thus an ℓ -cycle of $\rho_{a,b}$ is exactly one alternating 2ℓ -cycle of $M_a \cup M_b$. Cyclically permuting the three coordinate classes proves the column and symbol statements and the FFF equivalence.

A 4-cycle in $M_a \cup M_b$ is exactly a 2-cycle of $\rho_{a,b}$, hence a 2-by-2 Latin subsquare on two rows, two columns, and two symbols. This is an intercalate. The same intercalate is counted once by its row pair, once by its column pair, and once by its symbol pair, proving the three equalities. $\square$

Lemma 2.6 (Global three-view sign identity). *For every Latin square of order n ,*

$$R(L)C(L)S(L) = (-1)^{\binom{n}{2}}.$$

Proof. Give each coordinate copy of X the same fixed order, and let $\tau(x, y) = (y, x)$ on X^2 . The $\binom{n}{2}$ off-diagonal pairs are the transpositions of τ , so $\text{sgn}(\tau) = (-1)^{\binom{n}{2}}$.

The row-entry map $A(r, c) = (r, L(r, c))$ is block diagonal by rows and has sign $R(L)$. The map $B(r, c) = (c, L(r, c))$ is a coordinate transposition followed by the block-diagonal family of column permutations, so its sign is $(-1)^{\binom{n}{2}}C(L)$. Likewise, $D(c, s) = (s, s_s(c))$ has sign $(-1)^{\binom{n}{2}}S(L)$. Since $A = \tau \circ D \circ B$, taking signs gives

$$R(L) = (-1)^{3\binom{n}{2}}S(L)C(L) = (-1)^{\binom{n}{2}}S(L)C(L),$$

which is equivalent to the claimed identity. $\square$

For later comparison, let $C_{v,k}$ denote the total number of k -cycles among all induced line-pair permutations in view v .

Proposition 2.7 (Aggregate cycle-profile baseline). *Let n be even and let p be prime. If*

$$\sum_{k=2}^n a_k x_k + b = 0 \pmod{p}$$

for every cycle-count vector $(x_2, \dots, x_n)$ of a fixed-point-free permutation of degree n , then, separately for each view v ,

$$\sum_{k=2}^n a_k C_{v,k} + \binom{n}{2} b = 0 \pmod{p}.$$

Moreover,

$$C_{\text{row},2} = C_{\text{col},2} = C_{\text{sym},2}$$

and

$$\sum_{v \in \{\text{row}, \text{col}, \text{sym}\}} \sum_{\substack{2 \leq k \leq n \\ k \text{ even}}} C_{v,k} = \frac{n(n-1)}{2} \pmod{2}.$$

Proof. The aggregate profile in one view is the sum of the $\binom{n}{2}$ fixed-point-free line-pair profiles, proving the first identity. The second identity is the intercalate bijection from Proposition 2.5. For the last identity, the sign of a fixed-point-free permutation of even degree is $(-1)^e$, where e is its number of even cycles. Multiplying relative-permutation signs over all line pairs in a view gives the product of its line signs. Lemma 2.6 gives the product over the three views as $(-1)^{n(n-1)/2}$, which proves the congruence. $\square$

Proposition 2.8 (Line-pair sign cut). *Fix one view and write its line bijections as λ_i . For $P_{ij} = \lambda_j^{-1} \lambda_i$, the pairs for which P_{ij} is odd form a cut of the complete graph on the lines. If n is even and the view is F , then the pairs for which P_{ij} has an odd number of cycles form the same cut. Consequently, for even $n > 2$, no view can have every induced line-pair permutation equal to one n -cycle.*

Proof. Put $\varepsilon_i = \text{sgn}(\lambda_i)$. Since sign is a homomorphism,

$$\text{sgn}(P_{ij}) = \varepsilon_i \varepsilon_j.$$

Thus P_{ij} is odd exactly when the endpoint signs differ. These pairs are the edges of the cut between positive and negative lines.

In an F view of even degree, every cycle of P_{ij} has even length and therefore negative sign. Hence $\text{sgn}(P_{ij}) = (-1)^{c(P_{ij})}$, where $c(P_{ij})$ is the number of cycles. Finally, if all relative permutations were n -cycles, every edge would be negative. A triangle of three lines would then cross the sign cut three times, whereas every cut meets a cycle evenly. $\square$

An *isotopy* independently permutes rows, columns, and symbols. A *parastrophy* permutes the three coordinate roles. Their combined orbits are the *main classes* (also called species).

Proposition 2.9. *Under isotopy, each of the three view bits is invariant. Under parastrophy, the three bits are permuted. Consequently, being FFF and having at least one failing view are main-class invariants.*

Proof. Let an isotopy act by permutations α, β, γ of rows, columns, and symbols. The transformed row maps satisfy $r'_{\alpha(a)} = \gamma r_a \beta^{-1}$, and hence

$$(r'_{\alpha(b)})^{-1} r'_{\alpha(a)} = \beta (r_b^{-1} r_a) \beta^{-1}.$$

Thus row cycle lengths are unchanged. The column calculation conjugates by α , and the symbol calculation conjugates by β . Parastrophy only permutes which coordinate pair is called row, column, or symbol. $\square$

3 Elementary structural results

3.1 Odd orders and group tables

Proposition 3.1. *Every Latin square of odd order $n > 1$ has pattern TTT.*

Proof. By Lemma 2.1, every induced permutation between two distinct lines is fixed-point-free on an odd set. If all of its nontrivial cycles were even, their lengths would have even sum. Hence it has an odd cycle of length greater than one. This argument applies separately to any pair of distinct rows, columns, and symbols. $\square$

Let G be a finite group and let $L_G(x, y) = xy$ be its Cayley table.

Proposition 3.2. *The row permutation induced by rows a, b of L_G is left multiplication by $b^{-1}a$. The column permutation induced by columns c, d is right multiplication by cd^{-1} . The symbol permutation induced by symbols u, v is right multiplication by $u^{-1}v$. Each cycle length is therefore the order of the corresponding group element.*

Proof. The row identity is $r_b^{-1}r_a(x) = b^{-1}ax$. Likewise $q_d^{-1}q_c(x) = xcd^{-1}$. Finally, $s_u(c) = uc^{-1}$ and $s_v^{-1}(a) = a^{-1}v$, whence $s_v^{-1}s_u(c) = cu^{-1}v$. Left or right multiplication by g has orbits equal to cosets of $\langle g \rangle$, all of length $\text{ord}(g)$. $\square$

Corollary 3.3. *The Cayley table of a finite group G is FFF if and only if G is a 2-group.*

Proof. If G is a 2-group, every nonidentity element has even order, so Proposition 3.2 gives no nontrivial odd cycle. Conversely, if $|G|$ has an odd prime divisor, Cauchy's theorem supplies a nonidentity element of odd order; the corresponding induced permutations give a failing view. $\square$

3.2 A closure criterion for group isotopy

Fix a base row b of a Latin square L and define

$$H_L(b) = \{r_b^{-1}r_x : x \in X\} \subseteq \text{Sym}(X).$$

Proposition 3.4 (Basis-family closure criterion). *A Latin square L is isotopic to a group table if and only if $H_L(b)$ is closed under composition. This condition is independent of the chosen base row.*

Proof. Suppose first that $H = H_L(b)$ is closed. It contains the identity, has $|X|$ distinct elements, and is a finite submonoid of a symmetric group; hence it is a group. For each $z \in X$, the values $(r_b^{-1}r_x)(z)$, as x varies, are all distinct, since a column of L contains every symbol once. Thus H acts regularly on X .

Fix $z_0 \in X$. Relabel rows by $x \mapsto h_x = r_b^{-1}r_x$. Use the regular action to identify every column z with the unique $h_z \in H$ satisfying $h_z(z_0) = z$, and identify every relabeled symbol $r_b^{-1}(u)$ in the same way. The entry in row h_x and column h_z is represented by the unique element sending z_0 to $h_x h_z(z_0)$, namely $h_x h_z$. The resulting isotope is therefore the Cayley table of H .

Conversely, for a group table the family at base row b is the full left regular representation $\{\lambda_{b^{-1}x} : x \in G\}$ and is closed. Isotopy conjugates the family and relabels its members, preserving closure. The same argument for any base row proves base independence. $\square$

The same criterion can equivalently be written in the column or symbol view, by parastrophy.

4 Subsquare and congruence obstructions

The FFF property is hereditary for Latin subsquares in a stronger, componentwise sense.

Theorem 4.1 (Subsquare monotonicity). *If M is a Latin subsquare of L , then*

$$\text{pat}(M) \leq \text{pat}(L)$$

componentwise, with $F < T$.

Proof. Let the rows, columns, and symbols of M be R_0, C_0, S_0 . For $a, b \in R_0$, the ambient row permutation $r_b^{-1}r_a$ preserves C_0 : if $c \in C_0$, then $L(a, c) \in S_0$, and row b contains this symbol at a unique column of C_0 . Its restriction is precisely the induced row permutation of M . Thus every odd row cycle of M is also an odd row cycle of L . Interchanging rows and columns proves the column statement.

For $u, v \in S_0$ and $c \in C_0$, the row $s_u(c)$ lies in R_0 . In that row, the unique occurrence of v lies in a column $d \in C_0$. Hence $s_v^{-1}s_u$ preserves C_0 and restricts to the symbol permutation of M . This proves the symbol statement. $\square$

Corollary 4.2. *Every Latin subsquare of an FFF square is FFF. In particular, an FFF square contains no subsquare of odd order greater than one.*

Proof. The first assertion is Theorem 4.1 with $\text{pat}(L) = \text{FFF}$. The second follows from Proposition 3.1. $\square$

Together with Computational Theorem 6.1, the same implication excludes order-6 subsquares; that additional exclusion inherits the computational status of the order-6 theorem.

There is an exact linear-algebraic description of all half-order subsquares. Let M_L be the $3n$ -by- n^2 binary incidence matrix whose rows are the row, column, and symbol lines and whose columns are the occupied cells of L . Let

$$U = \{(\alpha \mathbf{1}_R, \beta \mathbf{1}_C, \gamma \mathbf{1}_S) : \alpha + \beta + \gamma = 0\} \subseteq \ker M_L^\top$$

and put

$$\delta_2(L) = (3n - 2) - \text{rank}_{\mathbb{F}_2} M_L.$$

Theorem 4.3 (Binary line kernel and half-order subsquares). *Individual embedded half-order Latin subsquares of L are in bijection with*

$$\ker M_L^\top \setminus U.$$

Equivalently, the nonzero classes in $\ker M_L^\top / U$ parametrize their complementary four-packs. Thus their number is exactly

$$4(2^{\delta_2(L)} - 1). \tag{1}$$

If $n = 2m$, where $m > 1$ is odd, then any one F -view forces

$$\text{rank}_{\mathbb{F}_2} M_L = 3n - 2. \tag{2}$$

In particular, (2) holds for every FFF square of order congruent to 2 modulo 4 and greater than 2.

Proof. A left-kernel vector is a triple (a, b, c) satisfying

$$a_r + b_j + c_{L(r,j)} = 0. \quad (3)$$

If one coordinate, say a , is constant, then fixing a column and varying its rows shows that c is constant; fixing a row then makes b constant. Hence every vector outside U has all three coordinates nonconstant.

Fix a row r . Since that row permutes the symbols, (3) identifies the multiset of c -values with that of $a_r + b$. Choosing rows with $a_r = 0$ and $a_r = 1$ gives

$$\text{wt}(c) = \text{wt}(b) = n - \text{wt}(b).$$

Thus $n = 2m$ and b, c each have weight m . The analogous fixed-column argument gives $\text{wt}(a) = m$. For their binary fibres A_x, B_y, C_z , equation (3) now gives

$$L(A_x, B_y) \subseteq C_{x+y} \quad (x, y \in \mathbb{F}_2).$$

Every set has size m ; Latinness makes each of the four displayed blocks an order- m Latin subsquare.

Adding an element of U complements zero or two of the three partitions and selects a different block in the same four-pack. Conversely, if $R_0 \times C_0$ is a half-order subsquare with symbol set S_0 , row and column counting forces the complementary pattern

$$R_0 \times C_0, R_1 \times C_1 \longrightarrow S_0, \quad R_0 \times C_1, R_1 \times C_0 \longrightarrow S_1.$$

The three complement indicators satisfy (3), and the chosen block recovers them uniquely. This proves both bijections. Since U has dimension two and the quotient has dimension $\delta_2(L)$, counting its nonzero classes proves the counting formula.

Finally suppose that $m > 1$ is odd and $\delta_2(L) > 0$. The preceding construction gives an order- m subsquare. Two distinct rows inside it induce a fixed-point-free permutation preserving its m columns. Such a permutation on an odd set has a nontrivial odd cycle, contradicting an F row view. The two parastrophes give the same conclusion for either other view, which proves the rank formula. The exception $m = 1$ is real: the order-2 FFF square has four singleton subsquares and $\delta_2 = 1$. $\square$

The complete finite audit independently constructs both sides of the bijection on 9778 tables and obtains exact set equality in every case. In particular, the 230 order-8 FFF representatives have

$$\delta_2 = 0, 1, 2, 3 \quad \text{with multiplicities} \quad 63, 156, 10, 1,$$

which explains their half-order-subsquare distribution $0^{63}4^{156}12^{10}28^1$. A hypothetical order-10 FFF square must have binary line rank 28 and no order-5 subsquare. This condition is not sufficient: all 135 tracked order-10 diagnostic tables already satisfy it, none is FFF, and that corpus is not exhaustive.

The same kernel equation has a uniform interpretation over every prime field, now in terms of quotient homotopies rather than individual subsquares. Moorhouse's character formula for loop-coordinatized 3-nets gives the general identity $\text{rank}_p M_L = 3n - 2 - \dim \text{Hom}(G, \mathbb{F}_p)$ and identifies its defect with elementary-abelian quotient characters [Moo91, Theorem 4.2]. We record a direct three-sorted proof in our notation because the individual cyclic quotient is the input needed for the FFF corollary below. The author's publication page flags an erratum for the original article; the reference entry therefore points to the author-hosted article copy and the separate erratum listing. The direct proof given here, rather than an unexamined appeal to the original pagination, is the logical basis for the specialization used below.

Theorem 4.4 (Prime-field line quotients). *Let p be prime and let $(a, b, c) \in \ker_{\mathbb{F}_p} M_L^\top$ be nonconstant modulo the two-dimensional constant subspace. Then $p \mid n$, each of a, b, c takes every value exactly n/p times, and their fibres satisfy*

$$L(R_x, C_y) \subseteq S_{-x-y} \quad (x, y \in \mathbb{F}_p).$$

After a suitable isotopy these three fibre partitions become one quasigroup congruence with p blocks of size n/p and quotient isotopic to C_p .

Over characteristic zero, M_L has rank $3n - 2$. Every prime dividing a nonzero Smith invariant factor of the integer matrix M_L divides n .

Proof. If one of a, b, c is constant, fixing a line in either other sort and using Latinness makes the remaining two constant. Let $\mu_b(t)$ and $\mu_c(t)$ be value multiplicities. Since every row permutes the symbols, equation (3), now read in $\mathbb{F}_p$, gives

$$\mu_c(t) = \mu_b(-a_r - t).$$

Comparing rows with $a_r - a_{r'} \neq 0$ makes μ_b invariant under a nonzero translation. That translation generates the additive group of $\mathbb{F}_p$, so μ_b is uniform. Thus $p \mid n$ and both b, c are balanced; the column argument balances a . The displayed fibre rule is the kernel equation itself.

Choose bijections from one partitioned n -set to the row, column, and symbol sets, matching equally labelled fibres. Transporting L by these three bijections gives an isotope whose block operation is $x * y = -x - y$ on $\mathbb{F}_p$. It is Latin and isotopic to addition, so block equivalence is compatible with multiplication and both divisions and is a quasigroup congruence.

In characteristic zero a finite nonempty multiset cannot be invariant under a nonzero translation, because its iterated translates would have infinite support. Hence only the two constant dependencies remain. Finally, if a prime divides a nonzero Smith invariant factor, reduction modulo that prime lowers the rank; the first part then forces the prime to divide n . $\square$

An eight-worker audit checks this theorem over $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7, \mathbb{F}_{11}$ on 9778 tables. It verifies all 1160 nonuniversal quotient classes and a separate implementation reconstructs 48890 modular ranks, with no errors.

We next use congruences in the quasigroup algebra with multiplication and the two division operations. All congruence blocks have equal size and form a Latin quotient.

Theorem 4.5 (Congruence obstructions). *Let L be a finite Latin square.*

- (i) *If L has a congruence with odd block size greater than one, then $\text{pat}(L) = \text{TTT}$.*
- (ii) *If the congruence blocks have size two and the quotient is Q , then $\text{pat}(L) = \text{pat}(Q)$ componentwise.*

Proof. For (i), choose distinct rows x, x' in one congruence block A and any column block B . The two left translations map B bijectively onto the same product block. Thus $L_{x'}^{-1} L_x$ preserves B and is fixed-point-free there. Since $|B|$ is odd, its restriction has a nontrivial odd cycle. Applying the same argument to the two relevant parastrophes proves the column and symbol bits.

For (ii), label every two-element block by $(i, 0), (i, 1)$, where $i \in Q$. The unique form of a binary Latin square gives a function $f: Q^2 \rightarrow \mathbb{F}_2$ such that

$$(i, a) * (j, b) = (i * j, a + b + f(i, j)).$$

Rows in the same quotient block induce only transpositions inside the column fibres. For rows in different quotient blocks, the lift above a quotient cycle of length ℓ is either two cycles of length ℓ or one cycle of length 2ℓ , according to its binary voltage. Thus an ambient odd cycle can occur only above an odd quotient cycle.

Conversely, suppose quotient rows i, i' induce an odd cycle C of length $\ell > 1$. Above a quotient column $j \in C$, the lifted row permutation comparing (i, a) with (i', a') has the form

$$(j, b) \mapsto (\pi(j), b + a + a' + f(i, j) + f(i', \pi(j))).$$

Its voltage around C is $a + a' + S$, because ℓ is odd, where S is independent of a, a' . Choose $a + a' = S$. The voltage is then zero, so the lift of C consists of two cycles of the same odd length ℓ . Hence a quotient row witness lifts to an ambient row witness. This proves equality of the row bits. The same complete argument in the parastrophes proves equality of the column and symbol bits. $\square$

Corollary 4.6 (Order- $2p$ simplicity). *Let p be an odd prime. Every quasigroup of order $2p$ with at least one F view is congruence-simple, and every isotope of it is congruence-simple.*

Proof. A nontrivial congruence has block-size/quotient-order pair $(p, 2)$ or $(2, p)$. The first is excluded by Theorem 4.5(i). In the second, the quotient has pattern TTT by Proposition 3.1, so (ii) excludes it. Both congruence cases force TTT, contradicting even one F view. Finally, the full view pattern is isotopy invariant by Proposition 2.9, and the same argument applies to every isotope. $\square$

Corollary 4.7 (Order- $2p$ line-incidence saturation). *Let $p > 1$ be an odd prime and let L have order $2p$. If even one view of L is F, then M_L has rank $6p - 2$ over every field, all its nonzero Smith invariant factors are 1, and*

$$\mathbb{Z}^{6p} / \text{im}_{\mathbb{Z}} M_L \cong \mathbb{Z}^2.$$

In particular, this holds for every hypothetical order-10 FFF square.

Proof. By Theorem 4.4, a torsion prime must divide $2p$. A characteristic-two defect yields, after isotopy, a congruence with two blocks of odd size p , so Theorem 4.5(i) gives pattern TTT. A characteristic- p defect yields blocks of size two; part (ii) identifies the pattern with that of the odd-order quotient, again TTT by Proposition 3.1. Isotopy invariance contradicts the assumed F view in either case. Thus no Smith prime remains, proving saturation and the displayed cokernel. $\square$

Thus a hypothetical order-10 FFF square must be isotopically congruence-simple. This removes every congruence-imprimitive quasigroup construction, but it does not rule out simple quasigroups and therefore does not decide order 10.

The binary equality iterates through congruence series.

Corollary 4.8 (Prime-factor congruence series). *Suppose that a quasigroup admits a chain of congruence quotients ending in the one-element quasigroup, with every congruence block size prime. It is FFF if all those primes are 2, and it has pattern TTT if at least one is odd.*

Proof. A binary step preserves the entire pattern by Theorem 4.5(ii). At an odd step, Theorem 4.5(i) gives TTT independently of the lower quotient. Iterating proves both statements. $\square$

This characterizes FFF within this congruence-decomposable class. It does not imply the former global power-of-two conjecture, which is refuted by the order-12 witness in Computational Theorem 7.1.

The same theorem also gives a census-free nongroup example.

Theorem 4.9 (Explicit nongroup FFF loop). *On $\mathbb{Z}_4 \times \mathbb{F}_2$, define*

$$(i, a) * (j, b) = (i + j, a + b + \mathbf{1}_{(i,j)=(1,1)}),$$

with coordinates taken modulo 4 and 2, respectively. This is an FFF loop of order 8 that is not group-isotopic.

Proof. The identity is $(0, 0)$, and the first-coordinate fibres form a congruence with quotient C_4 . The quotient is FFF by Corollary 3.3, so binary pattern equality makes the loop FFF. It is not associative because

$$((1, 0) * (1, 0)) * (2, 0) = (0, 1), \quad (1, 0) * ((1, 0) * (2, 0)) = (0, 0).$$

At the identity base row, the basis family in Proposition 3.4 is the family of left translations. Closure would force $L_x L_y = L_{x*y}$ and hence associativity. The displayed failure is therefore a closure failure, so the loop is not group-isotopic. $\square$

5 Direct products

Let L be a Latin square on X and K a Latin square on Y . Their direct product $P = L \times K$ is the Latin square on $X \times Y$ defined by

$$P((x, y), (x', y')) = (L(x, x'), K(y, y')).$$

Lemma 5.1 (Product-cycle formula). *Let π and τ be permutations. If x lies on a π -cycle of length m and y lies on a τ -cycle of length n , then (x, y) lies on a cycle of length $\text{lcm}(m, n)$ under $\pi \times \tau$.*

Proof. The pair returns after $t > 0$ steps exactly when both $m \mid t$ and $n \mid t$. The least such t is $\text{lcm}(m, n)$. $\square$

Theorem 5.2 (Pattern product theorem). *For arbitrary Latin squares L and K ,*

$$\text{pat}(L \times K) = \text{pat}(L) \text{ OR } \text{pat}(K)$$

componentwise.

Proof. Rows of P are indexed by (a, α) , and their row maps are $r_{(a, \alpha)}^P = r_a^L \times r_\alpha^K$. Hence the induced permutation between (a, α) and (b, β) is

$$\rho_{(a, \alpha), (b, \beta)}^P = \rho_{a, b}^L \times \rho_{\alpha, \beta}^K.$$

If one coordinate of the two product rows is equal, the corresponding factor is the identity. The identical calculation for column maps gives $\kappa^P = \kappa^L \times \kappa^K$.

For the symbol view,

$$s_{(u, \eta)}^P(c, d) = (s_u^L(c), s_\eta^K(d)),$$

and therefore $\sigma^P = \sigma^L \times \sigma^K$. Thus the same argument applies in all three views.

If a factor has a nontrivial odd cycle in one view, keep the other factor line fixed. The identity contributes cycle length 1, so the product contains the same odd cycle. Conversely, suppose a product induced permutation has an odd cycle of length $\ell > 1$. By Lemma 5.1, $\ell = \text{lcm}(m, n)$ for factor cycle lengths m, n . Both are odd, and at least one is greater than one. That factor already has a nontrivial odd cycle in the same view. This proves the Boolean OR formula in each view. $\square$

Corollary 5.3. *$L \times K$ is FFF if and only if both L and K are FFF. In particular, product with a finite 2-group table preserves the full pattern of the other factor.*

Proof. The first assertion is the all- F case of Theorem 5.2. For the second, combine the same theorem with Corollary 3.3. $\square$

Theorem 5.4 (Group-isotopy product theorem). *$L \times K$ is group-isotopic if and only if both L and K are group-isotopic.*

Proof. At a product base row (b, c) , the basis family factors as

$$H_{L \times K}(b, c) = H_L(b) \times H_K(c).$$

If both factor families are closed, their Cartesian product is closed. Conversely, both factor families contain the identity. If the product family is closed, then $(h_1, \text{id})(h_2, \text{id})$ lies in it for all $h_1, h_2 \in H_L(b)$, proving closure of $H_L(b)$; using (id, k_i) proves closure of $H_K(c)$. Apply Proposition 3.4. $\square$

Corollary 5.5. *Let L be a non-group-isotopic FFF Latin square of order n . For every $m \geq 0$, the product $L \times C_2^m$ is a non-group-isotopic FFF Latin square of order $n2^m$.*

Proof. The table of C_2^m is FFF by Corollary 3.3; apply Corollary 5.3 and Theorem 5.4. $\square$

6 Computational results

The following statements are exhaustive finite computations. Their frozen inputs, programs, commands, outputs, and hashes are specified in Section 12.

Computational Theorem 6.1 (Small orders). *The unique reduced Latin square of order 2 is FFF, and all four reduced Latin squares of order 4 are FFF. None of the 9408 reduced Latin squares of order 6 is FFF. More precisely, their patterns are*

$$6600 \text{ TTT}, \quad 936 \text{ TFF}, \quad 936 \text{ FTF}, \quad 936 \text{ FFT}.$$

The exact-cover generator loads no cached square list. Every generated square was checked to be Latin and reduced, all 9408 table strings were distinct, and the generated set agrees exactly with the ANU reduced-order-6 data file [McK]. Proposition 2.9 therefore extends nonexistence from reduced squares to all Latin squares of order 6.

Computational Theorem 6.2 (Order-8 pattern census). *Across the 283657 order-8 main-class representatives, the pattern distribution is*

Pattern	Count	Pattern	Count	Pattern	Count	Pattern	Count
FFF	230	FFT	81	FTF	40	FTT	315
TFF	133	TFT	209	TTF	232	TTT	282417

Among the 230 FFF main classes, exactly 5 are group-isotopic and 225 are not group-isotopic.

Computational Theorem 6.3 (Subsquare census of the FFF classes). *Among the 230 FFF representatives, the number of intercalates ranges from 2 to 112. Each endpoint is attained by exactly one representative: local source line 10776 (ANU record 10775) is the unique minimum class and is not group-isotopic, while local line 1 (ANU record 0) is the unique maximum class and is group-isotopic. Local physical lines are numbered from one; ANU records are numbered from zero. No representative contains an order-3 subsquare. The distribution of the number of order-4 subsquares is*

$$0^{63}, \quad 4^{156}, \quad 12^{10}, \quad 28^1,$$

where exponents count representatives.

The subsquare validator finds no violation of Theorem 4.1. The multiples of four in the last distribution also follow directly: an order- m subsquare inside an order- $2m$ square forces the three complementary blocks to be Latin subsquares, so half-order subsquares occur in complementary quartets.

The ordered non-symmetric labels refer to the representatives in the cited ANU file. Parastrophy permutes the three bits, so labels such as FFT are not themselves main-class invariants. The FFF count, the existence of every pattern, and the group-isotopy split are unaffected.

The scan tests every unordered pair of lines in all three views. A separate closure-family computation reproduces the 5/225 split using Proposition 3.4. The frozen ANU input is an exact byte match to the file served by the cited URL on August 9, 2026.

Corollary 6.4. *For every $m \geq 3$, Latin squares of order 2^m realize all eight patterns. Every such order also contains a non-group-isotopic FFF Latin square.*

Proof. Computational Theorem 6.2 supplies one order-8 square of each pattern. Take direct products with the FFF group table of C_2^{m-3} and use Theorem 5.2. For the second assertion, start from the explicit loop in Theorem 4.9; non-group-isotopy is preserved by Theorem 5.4. $\square$

This is an existence result only. It neither counts higher-order main classes nor asserts that products of distinct order-8 classes remain distinct.

7 Explicit non-power-of-two witnesses and a dyadic lift bound

The computational nonexistence result at order 10 does not extend to all even non-powers of two. The following finite claims are checked by direct three-view cycle scans, not by an uncompleted solver search.

Computational Theorem 7.1 (Explicit non-power-of-two witnesses). *There are FFF Latin squares of orders 12, 14, and 36. The displayed order-12 table is reduced. Thus the former conjecture that FFF squares of order greater than one exist only at powers of two is false.*

L_{12}	0	1	2	3	4	5	6	7	8	9	10	11
0	0	1	2	3	4	5	6	7	8	9	10	11
1	1	0	3	2	5	4	7	6	10	11	8	9
2	2	3	0	1	7	6	5	4	9	8	11	10
3	3	2	1	0	6	7	4	5	11	10	9	8
4	4	7	5	6	8	11	9	10	3	0	2	1
5	5	6	4	7	9	10	8	11	2	1	3	0
6	6	5	7	4	10	9	11	8	1	2	0	3
7	7	4	6	5	11	8	10	9	0	3	1	2
8	8	9	10	11	3	2	1	0	6	7	4	5
9	9	8	11	10	0	1	2	3	4	5	6	7
10	10	11	8	9	2	3	0	1	5	4	7	6
11	11	10	9	8	1	0	3	2	7	6	5	4

Table 1: An explicit reduced order-12 FFF witness. Every row and column is a permutation; all 198 induced line-pair permutations across the three views have only even cycles.

For odd $m > 1$, let $h(m)$ be the least $a \geq 0$ for which an FFF square of order $m2^a$ exists. The odd-order obstruction, the order-6 computation, and Computational Theorem 7.1 give $h(3) = 2$ and $h(7) = 1$; the order-36 table gives $h(9) \leq 2$. Nothing here decides existence at order 18.

The classical polygon-and-centre round-robin factorisation gives a separate, particularly simple family. Cheng and Sgueglia [CS26, Introduction] recall this factorisation and its prime-order perfectness; the following three-view FFF criterion is proved directly, without claiming priority for the construction or an exhaustive literature classification.

Theorem 7.2 (Prime round-robin family). *Let p be an odd prime and $X = \mathbb{F}_p \cup \{\infty\}$. Define M by $M(\infty, \infty) = \infty$, $M(\infty, x) = M(x, \infty) = x$ for finite x , $M(x, x) = \infty$ for finite x , and $M(x, y) = (x + y)/2$ for distinct finite x, y . Then M is Latin and is FFF if and only if $p \equiv 3 \pmod{8}$.*

Proof. In each finite row a , the affine map $y \mapsto (a + y)/2$ takes the value a only at $y = a$. Replacing that entry with ∞ and placing a at column ∞ makes the row a permutation. The infinity row is the identity. Symmetry gives the Latin property for columns as well.

Write R_a for a row permutation and $d = \text{ord}_p(2)$. R_∞ is the identity, while R_a swaps a and ∞ and acts, after translating a to zero, as multiplication by $1/2$ on $\mathbb{F}_p^*$. Hence an infinity/finite row pair has type $2d^{(p-1)/d}$. For distinct finite a, b , put $P = R_b^{-1}R_a$. Apart from a , $2b - a$, and ∞ , the map is translation $x \mapsto x + a - b$. At the exceptional inputs,

$$P(a) = b, \quad P(2b - a) = \infty, \quad P(\infty) = 2a - b,$$

and $P(b) = a$. In the single p -cycle of the translation, the segment $2b - a \rightarrow b \rightarrow a \rightarrow 2a - b$ is replaced by the 2-cycle (a, b) and the connection $2b - a \rightarrow \infty \rightarrow 2a - b$. The remaining cycle has length $p - 1$, including when $p = 3$. Thus all finite/finite row pairs have type $(2, p - 1)$, and the row view is F exactly when d is even. Symmetry gives the identical column condition.

For a symbol s , let T_s send a column to the row containing s in it. Then T_∞ is the identity; for finite s ,

$$T_s(\infty) = s, \quad T_s(s) = \infty, \quad T_s(x) = 2s - x \quad (x \in \mathbb{F}_p \setminus \{s\}).$$

Thus each T_s is a fixed-point-free involution. An infinity/finite symbol pair has type $2^{(p+1)/2}$. Affine relabelling of the finite points reduces any two distinct finite symbols to $0, 1$. The relative permutation T_1T_0 has precisely the cycles

$$(0, 1, 3, 5, \dots, p - 2), \quad (\infty, 2, 4, 6, \dots, p - 1),$$

both of length $(p+1)/2$. Hence the symbol view is F exactly when $p \equiv 3 \pmod{4}$.

For $p \equiv 7 \pmod{8}$, the supplement $(2/p) = 1$ and Euler's criterion make d divide the odd number $(p-1)/2$, so the row and column views are T. For $p \equiv 3 \pmod{8}$, $(2/p) = -1$ makes d even; the symbol view is also F. Primes $p \equiv 1 \pmod{4}$ already fail in the symbol view. This proves the equivalence. $\square$

By Dirichlet's theorem, infinitely many primes satisfy $p \equiv 3 \pmod{8}$; apart from $p = 3$, their orders $p+1$ are not powers of two. At $p = 19$ the table has order 20. Together with the odd-order obstruction Proposition 3.1 and the separately audited computational exclusion of order 10, this gives the sharp threshold $h(5) = 2$. This last equality depends on the exact order-10 computation, whereas Theorem 7.2 is algebraic.

A second class at order 20. The classical no-hexagon STS(19), with explicit starters in Erskine–Griggs [EG24, Section 3.2] and classification context in Colbourn et al. [CFG⁺10], yields a Steiner loop whose 570 two-line permutations have only even cycles. Its three-view cycle profile has no 18-cycle, whereas the midpoint table at $p = 19$ has type $(2, 18)$ in every row pair. Since the combined cycle-type profile is invariant under paratopy, these two examples are not in the same main class. This initial two-class control does not reproduce the external STS classification. The separate trade-family audit in Section 10 strengthens the represented class lower bound to 514.

Theorem 7.3 (Three-point lift bound). *For every odd integer $m > 1$, there is an FFF Latin square of order $m2^{\text{ord}_m(2)}$. Consequently*

$$h(m) \leq \text{ord}_m(2),$$

and FFF squares exist at every order $m2^a$ with $a \geq \text{ord}_m(2)$.

Proof. The case $m = 3$ is supplied by the order-12 table above, since $\text{ord}_3(2) = 2$. Suppose $m > 3$. Set $e = \text{ord}_m(2)$, take $K = \mathbb{F}_{2^e}$, and choose $r \in K^\times$ of multiplicative order m . Its existence follows from $m \mid (2^e - 1)$ and cyclicity of $K^\times$.

For coefficients $s_i \in K$, $i \in \mathbb{Z}_m$, define a multiplication on $\mathbb{Z}_m \times K$ by

$$(i, a) * (j, b) = (i + j, a + r^i b + s_i r^j). \quad (4)$$

The quotient coordinate and the nonzero coefficient r^i make this a Latin square. Put $t_i = r^{-i} s_i$ and regard t_i as a function $t(r^i)$ on $H = \langle r \rangle$. For a coset C of any nontrivial subgroup of H , write

$$S_C = \sum_{x \in C} x t(x), \quad T_C = \sum_{x \in C} x^{-1} t(x).$$

We first verify that pairwise distinct t_i and nonzero S_C, T_C for all such cosets imply FFF in all three views.

Two distinct lines in the same quotient fiber have relative permutations that are nonzero translations of K : for rows the translation is $r^{-i}(a + a')$, for columns it is $r^i(b + b')$, and for symbols it is $r^{-(k-j)}(u + u')$ in each column fiber. Each is an involution without fixed points. For distinct quotient rows i, k , put $d = i - k$ and let ℓ be the order of d in $\mathbb{Z}_m$. Directly composing their row maps gives

$$R_{(k, a')}^{-1} R_{(i, a)}(j, b) = (j + d, r^d b + r^{-k}(a + a') + r^{j-k}(s_i + r^d s_k)).$$

The quotient shifts by d ; its ℓ -step fiber return is

$$b \mapsto b + r^{j-i}(s_i + r^d s_k) = b + r^j(t_i + t_k).$$

For distinct quotient columns j, l , the quotient shift is $j - l$ and the one-step map, with $d = j - l$, is

$$C_{(l,b')}^{-1} C_{(j,b)}(i, a) = (i + d, a + r^i(b + b') + r^j s_i + r^l s_{i+d}).$$

Its return on the associated coset C is

$$a \mapsto a + (r^j + r^l) S_C.$$

For distinct quotient symbols $(k, u), (l, v)$, use the symbol-line map

$$S_{(k,u)}(j, b) = (k - j, u + r^{k-j} b + s_{k-j} r^j).$$

With $d = l - k$, one step of $S_{(l,v)}^{-1} S_{(k,u)}$ is

$$(j, b) \mapsto (j + d, b + r^{j-k}(u + v) + (1 + r^d) r^j t_{k-j}).$$

Summing over a quotient cycle gives the return

$$b \mapsto b + (1 + r^{l-k}) r^k T_C.$$

In each calculation the line-label terms vanish because a nontrivial quotient cycle has odd length $\ell > 1$ and the relevant geometric sum of an ℓ th root other than one is zero. Every displayed multiplier outside S_C or T_C is nonzero. A nonzero characteristic-two translation has order two, so each lifted cycle has even length 2ℓ . This establishes the asserted three-view criterion.

It remains to choose the coefficients. Put $D = r + 1$, $u = D^{-1}$, and start with the injective function $t_0(1) = 1$ and $t_0(x) = (x + 1)^{-1}$ for $x \in H \setminus \{1\}$. Its values at $1, r, r^2$ are $1, u, u^2$, which are distinct for $m > 3$. Cyclically permute these three values:

$$t(1) = u, \quad t(r) = u^2, \quad t(r^2) = 1, \quad t(x) = (x + 1)^{-1} \quad (x \notin \{1, r, r^2\}),$$

and set $s_i = r^i t(r^i)$. Injectivity is retained.

Let $C = aJ$ be a coset of a subgroup J of odd order $\ell > 1$, and write $c = a^\ell$. If $1 \notin C$, then $\prod_{x \in C} (X + x) = X^\ell + c$ and logarithmic differentiation at $X = 1$ gives $\sum_{x \in C} (1 + x)^{-1} = (1 + c)^{-1}$. Also $\sum_{x \in C} x^{-1} = 0$. Any coset untouched by the patch therefore has

$$S_C = \frac{c}{1 + c}, \quad T_C = \frac{1}{1 + c},$$

both nonzero.

Now let J be proper, set $z = r^\ell$, and let $A = (\ell - 1)/2 \pmod{2}$, interpreted as an element of K . Its three affected cosets J, rJ, r^2J are distinct. The value changes relative to t_0 are $u + 1$, r/D^2 , and r^2/D^2 , respectively. Pairing x with x^{-1} on $J \setminus \{1\}$ gives $\sum_{x \in J \setminus \{1\}} (1 + x)^{-1} = A$. Including $t_0(1) = 1$, the unpatched moments on J are $S_J(t_0) = A + 1$ and $T_J(t_0) = A$. The other two cosets use the preceding unpatched formula. Adding the appropriate value changes yields

C	S_C	T_C
J	$A + u$	$A + 1 + u$
rJ	$\frac{z + r^2}{(1 + z)D^2}$	$\frac{z + r^2}{(1 + z)D^2}$
r^2J	$\frac{[z(r^2 + r + 1) + r^2]^2}{(1 + z^2)D^2}$	$\frac{z^2 + r^2}{(1 + z^2)D^2}$

All denominators are nonzero. The J entries are nonzero because $u \notin \{0, 1\}$. The rJ numerator could vanish only if $\ell \equiv 2 \pmod{m}$, and the last T numerator only if $\ell \equiv 1 \pmod{m}$; neither is possible for a proper divisor $1 < \ell < m$. The only possible failure is therefore

$$r^\ell(r^2 + r + 1) = r^2. \quad (5)$$

At most one exponent modulo m satisfies this equation. If it holds for a proper divisor ℓ , replace r by r^{-1} . A bad proper divisor j for the inverse would require $r^2 + r + 1 = r^j = r^{2-\ell}$, so $j = m + 2 - \ell > m/2$. Every proper divisor of odd m is at most $m/3$; there is no such j . Thus either r or its inverse makes every proper coset moment nonzero.

For the full-group coset H , all three patched points lie together. With $A = (m - 1)/2 \pmod{2}$, direct summation gives

$$S_H = 1 + A + r^2 + \frac{r}{r+1}, \quad T_H = A + \frac{r}{r+1}.$$

When $m \equiv 3 \pmod{4}$, these are $r(r^2 + r + 1)/(r + 1)$ and $1/(r + 1)$; the first could vanish only if r had order three. When $m \equiv 1 \pmod{4}$, they are $(r^3 + r^2 + 1)/(r + 1)$ and $r/(r + 1)$. The cubic is irreducible over $\mathbb{F}_2$ and its roots have order seven, incompatible with $m \equiv 1 \pmod{4}$. Hence both moments are nonzero. The three-view criterion proves that (4) is FFF.

Finally, multiplying by the table of an elementary abelian 2-group preserves FFF by Corollary 5.3, giving every exponent $a \geq e$. $\square$

Remark 7.4 (The exceptional affine base case). At $m = 3$ the affine formula (4) cannot be FFF, although Table 1 gives an order-12 FFF square. Put $t_i = r^{-i}s_i$. Row-F requires distinct t_i ; column-F and symbol-F require $S = t_0 + rt_1 + r^2t_2 \neq 0$ and $T = t_0 + r^2t_1 + rt_2 \neq 0$. Common translation and nonzero scaling give $(t_0, t_1) = (0, 1)$. Since $r^2 + r + 1 = 0$, $t_2 = r$ forces $T = 0$, whereas $t_2 = r^2$ forces $S = 0$.

The finite checks in the associated companion run evaluate all subgroup coset moments for $m = 5, 7, 9, 15, 21$ and all line-pair cycles at orders 56, 80, and 240. They control the implementation; the universal quantifier in Theorem 7.3 rests on the argument above. The theorem does not determine $h(m)$ in general, resolve order 18, classify higher-order main classes, or establish literature priority.

8 A common binary fiber over prime-square quotients

The preceding bound may require a larger binary exponent when the odd part is a prime square. The following construction uses a single binary fiber over a two-dimensional odd quotient. It is an internally audited research result awaiting independent mathematical review, not a claim of literature priority.

Theorem 8.1 (Quadratic-norm prime-square lift). *Let p be an odd prime, $e = \text{ord}_p(2)$, and D a positive multiple of e . If $q = 2^D > 3(p^2 - 1)$, then there is an FFF Latin square of order p^2q . Every relative two-line cycle in every view has length 2 or $2p$. In particular, $2^e > 3(p^2 - 1)$ implies $h(p^2) \leq e$.*

Proof. Let $G = \mathbb{F}_p^2$, choose a nonsquare $\nu \in \mathbb{F}_p$, and put $B(x, y) = x_1y_1 - \nu x_2y_2$ and $Q(x) = B(x, x)$. The norm Q is anisotropic: if $Q(d) = 0$ and $d_2 \neq 0$, then $\nu = (d_1/d_2)^2$, a contradiction; the case $d_2 = 0$ is immediate. Choose ω of order p in $K = \mathbb{F}_q$ and write $\chi(t) = \omega^t$. For a twist array $c : G \times G \rightarrow K$, define

$$(x, a) * (y, b) = (x + y, \chi(2B(x, y))a + \chi(Q(x))b + c(x, y)). \quad (6)$$

Both fiber coefficients are nonzero, so this is Latin for every c .

We calculate the relative action of two lines with distinct quotient coordinates. For rows x, z , put $d = x - z$ and $y_t = y + td$. For columns y, z , put $d = y - z$ and $x_t = x + td$. For symbols k, l , put $d = l - k$, $y_t = y + td$, and $x_t = k - y_t$. In each case the quotient orbit has length p . Solving (6) and then the inverse line map gives a fiber step

$$w_{t+1} = \chi(D_t)w_t + \chi(E_t)a + \chi(F_t)b + \chi(U_t)c(C_{1,t}) + \chi(V_t)c(C_{2,t}).$$

The slope exponents and the two twist cells are

$$\begin{aligned} D_t^{\text{row}} &= Q(x) - Q(z), & (C_{1,t}, C_{2,t}) &= ((x, y_t), (z, y_t + d)), \\ D_t^{\text{col}} &= 2B(x_t, y) - 2B(x_t + d, z), & (C_{1,t}, C_{2,t}) &= ((x_t, y), (x_t + d, z)), \\ D_t^{\text{sym}} &= 2B(x_t, d), & (C_{1,t}, C_{2,t}) &= ((x_t, y_t), (x_t, y_t + d)). \end{aligned}$$

Every $\chi(U_t)$ and $\chi(V_t)$ is nonzero. For clarity in the symbol case, the line map at symbol (k, a) sends a column (y, w) to the unique row $(k - y, u)$, where

$$u = \chi(Q(k - y) - 2B(k - y, y))w + \chi(-2B(k - y, y))(a + c(k - y, y)).$$

Taking the inverse of the second symbol map yields the third line above.

The sum of the p slope exponents is zero in each view: it is constant for rows and affine in t for columns and symbols, and $\sum_{t \in \mathbb{F}_p} t = 0$. The exponent of a label contribution from step t gains the later slopes $\sum_{s > t} D_s$. Expanding the quadratic form gives the following two accumulated exponents, where x, y denote the starting quotient row and column in the respective line:

	first line label	second line label
row	$2B(x, y) - Q(x) + tQ(d)$	$2B(z, y + d) - Q(x) - tQ(d)$
col	$Q(x) - 2B(x, y) - tQ(d)$	$Q(x) - 2B(x, z) + (t + 1)Q(d)$
sym	$-Q(x) - tQ(d)$	$-Q(x) - 2B(x, d) + tQ(d)$

Since $Q(d) \neq 0$, summing the character of any one of these affine progressions over $t \in \mathbb{F}_p$ gives zero. Thus both line-label coefficients cancel, the p -step slope is one, and the fiber return is $w \mapsto w + \Lambda_C(c)$ for a linear form in the twist cells.

The two cells at each step lie on different quotient lines. Within a line the p visited cells are distinct, also for symbols because a cell has a unique quotient symbol. Thus each Λ_C contains exactly $2p$ distinct variables with nonzero coefficients. In particular it is not the zero form. A fixed cell belongs to exactly $p^2 - 1$ such forms in each of the three views: one quotient orbit for each other quotient line. Its total incidence is $3(p^2 - 1)$.

Order the p^4 cells. Enforce $\Lambda_C(c) \neq 0$ when the last cell in its support is assigned. Previously assigned cells then forbid precisely one value for this cell. At most $3(p^2 - 1)$ conditions close at any cell, so $q > 3(p^2 - 1)$ leaves an allowed value. Induction makes every return a nonzero translation of K . Consequently different quotient lines have cycles of length $2p$. Two distinct lines in the same quotient fiber induce a nonzero translation on every complementary fiber, giving only 2-cycles. The argument applies to rows, columns, and symbols, proving FFF. $\square$

Corollary 8.2. *For all but $O((\log X)^2)$ primes $p \leq X$, with $e = \text{ord}_p(2)$, one has $h(p^r) \leq \lceil r/2 \rceil e$ for every $r \geq 1$.*

Proof. Theorem 8.1 supplies the prime-square factor; the order- $p2^e$ instance of Theorem 7.3 supplies an odd residual factor. Direct products preserve FFF, giving the stated bound

for each r . An exceptional prime satisfies $2^e \leq 3(p^2 - 1) < 4p^2$, hence $e < 2\log_2 p + 2$. If $p \leq X$ and $N = \lfloor 2\log_2 X + 2 \rfloor$, then p divides $\prod_{j=1}^N (2^j - 1)$. This product has fewer than $\sum_{j=1}^N j = N(N+1)/2$ distinct prime factors. $\square$

At $p = 11$, $e = 10$ and $3(p^2 - 1) = 360 < 1024$, giving an order-123904 construction. A separate standard-library checker validates all 239580 quotient-return conditions of a frozen compact twist certificate; it does not materialize that Latin table. Fresh smaller controls cover $p = 3, q = 64$ and $p = 5, q = 256$. The finite checks support the computation, whereas the universal quantifiers rest on the proof above. No claim is made about minimal exponents, order 18, or literature priority.

9 Affine prime-successor loops

For an odd prime p and $a \in \mathbb{F}_p \setminus \{0, 1\}$, put $b = 1 - a$. Define $Q_p(a)$ on $\mathbb{F}_p \cup \{\infty\}$ by

$$\infty * x = x * \infty = x, \quad x * x = \infty \ (x \in \mathbb{F}_p), \quad x * y = ax + by \ (x \neq y, \ x, y \in \mathbb{F}_p).$$

In each finite row or column, the affine bijection fixes the line label. Replacing that value by ∞ and placing the old label in the new position proves that this is a Latin loop. Write $[u]_p$ for the least positive residue of a nonzero field element.

Theorem 9.1 (Three-view affine criterion). *The loop $Q_p(a)$ is FFF if and only if*

$$\begin{aligned} [a]_p, [a^{-1}]_p, [b^{-1}]_p \text{ are even,} \\ \text{ord}_p(a), \text{ord}_p(b), \text{ord}_p(-b/a) \text{ are even.} \end{aligned} \tag{7}$$

Proof. Use a generic row parameter t , with $c = 1 - t$. A finite row relative to the identity row consists of a two-cycle and $(p-1)/\text{ord}_p(c)$ cycles of length $\text{ord}_p(c)$. Translations and nonzero scalings of finite labels are automorphisms and are two-transitive on them. It remains to compare rows 0, 1. Their relative map $P = r_1^{-1}r_0$, with $D = -t/c$, satisfies

$$P(\infty) = D, \quad P(0) = 1, \quad P(c^{-1}) = \infty, \quad P(y) = y + D \quad \text{at every other finite } y.$$

Number the translation cycle by jD , $j \in \mathbb{Z}_p$, and put $k = [-t^{-1}]_p$. Then $c^{-1} = kD$ and $1 = (k+1)D$. The redirected arrows give cycles of lengths $p-k$ and $k+1$, both even exactly when $[t^{-1}]_p$ is even.

The column view has parameter b . A finite symbol z , viewed as a map from columns to rows, swaps z, ∞ and otherwise sends y to $a^{-1}z - (b/a)y$. Its parameter is a^{-1} and its other coefficient is $-b/a$; the infinity symbol line is the identity. Applying the row calculation to these three parameters proves (7) in both directions. $\square$

Computational Theorem 9.2 (A residue-class boundary control). *The explicitly defined loop $Q_{97}(36)$ is FFF of order 98. Thus $n \equiv 2 \pmod{8}$, $n > 2$, is not a universal FFF obstruction.*

Indeed, $b = 62$, $36^{-1} = 62$, $62^{-1} = 36$, $-62/36 = 36$ in $\mathbb{F}_{97}$, and $36^3 = 62^3 = -1$. The orders and the three least residues in (7) are even. Primality and these identities are exact finite checks. The materialized table also passes all $3\binom{98}{2} = 14259$ line-pair checks. This is not an order-18 construction.

9.1 A uniform large-prime bound

Theorem 9.3 (Eventual prime-successor existence). *For every prime $p \geq 10^{12}$, more than $p/1000$ parameters produce FFF loops $Q_p(a)$.*

The analytic input is Katz's published mixed-character-sum estimate [Kat08, Theorem 1.5]. Its application and the explicit cutoff are checked below, not inferred from finite tables.

Proof. Put $U = \mathbb{F}_p \setminus \{0, 1\}$. For $p \equiv 3 \pmod{4}$, take the quadratic character χ and the sufficient mask

$$W(a) = \frac{1}{4}(1 - \chi(a))(1 - \chi(1 - a)).$$

For $p \equiv 1 \pmod{4}$, take a character of exact order four and use

$$W(a) = \frac{1}{8}(1 - \chi(a)^2)(1 - \chi(1 - a) + \chi(1 - a)^2 - \chi(1 - a)^3).$$

These are zero-one indicators. In the first case $a, b, -b/a$ are nonsquares. In the second, write $p - 1 = 2^s m$ with m odd. The primitive-root exponent of a is odd, that of b is 2 modulo 4, and that of $-b/a$ is odd. All three multiplicative orders are therefore even in either case. The mask's character expansion has coefficient absolute sum one and constant coefficient at least $1/8$.

Let $\psi(x) = \exp(2\pi i x/p)$ and let χ have order $q \in \{2, 4\}$. Except for the all-zero tuple, the sums

$$S(j, k; u, v, w) = \sum_{a \in U} \chi(a)^j \chi(1 - a)^k \psi(ua + v/a + w/(1 - a)), \quad 0 \leq j, k < q,$$

have absolute value at most $4\sqrt{p}$. For $u \neq 0$, apply the cited theorem with polynomial ua , rational remainder $v/a + w/(1 - a)$, and multiplicative polynomial $a^j(1 - a)^k$. All finite poles are simple; the pole order at infinity is below one, and every nonpole zero has nontrivial character multiplicity. With t endpoint poles and e other endpoint zeros, the estimate is $(2t + e)\sqrt{p}$. Deleting the remaining $2 - t - e$ endpoints changes the sum by at most that number, so the result is at most $4\sqrt{p}$. For $u = 0, v \neq 0$, substitute $z = 1/a$; for $u = v = 0, w \neq 0$, substitute $z = 1/(1 - a)$. Each is a bijection on U , gives a nonzero linear phase plus at most one simple finite pole, and changes the multiplicative factor to powers of $z, z - 1$, up to a constant. The same argument applies. With zero additive phase, the nontrivial multiplicative sums are Jacobi sums or one-character endpoint sums, bounded by $\sqrt{p}$. The all-zero tuple has sum $p - 2$.

Let $E(x)$ indicate nonzero even least residues. Geometric summation gives its normalized Fourier coefficients

$$\widehat{E}(0) = \frac{p - 1}{2p}, \quad |\widehat{E}(h)| = \frac{1}{2p|\cos(\pi h/p)|} \quad (h \neq 0).$$

Pairing $h, p - h$ and using $\sin x \geq 2x/\pi$ on $[0, \pi/2]$ bounds the nonzero coefficient sum by $\sum_{r=0}^{(p-3)/2} (2r + 1)^{-1} \leq 1 + \frac{1}{2} \log p$. The full absolute sum is at most $L(p) = 3/2 + \frac{1}{2} \log p$. Expanding $W(a)E(a)E(a^{-1})E((1 - a)^{-1})$ therefore bounds its parameter count by

$$N_p \geq \frac{(p - 2)(p - 1)^3}{64p^3} - 4\sqrt{p}L(p)^3. \quad (8)$$

The first term is at least $(p - 5)/64 \geq p/65$ for $p \geq 325$. The function $\sqrt{x}/L(x)^3$ increases once $L(x) > 3$. At $x = 10^{12}$, $\log 10 < 2.303$ gives $L(x) < 15.318$ and $260(15.318)^3 < 935000 < 10^6$, exact rational inequalities. It follows that $N_p > (1 - 0.935)p/65 = p/1000$. For example, the finite rational inequality $\sum_{k=0}^9 (2303/1000)^k/k! > 10$ proves the logarithm bound. $\square$

For $p > 3$ these loops are not group-isotopic. Associativity at $(0, 0, 1)$ would force $b^2 = 1$, hence $b = -1$, and at $(1, 0, 0)$ would force $a = -1$, contradicting $a + b = 1$. A group-isotopic loop is a group by the basis-family closure criterion: since its identity row is the identity permutation, closure gives $L_x L_y = L_{xy}$ after evaluating at the identity, hence associativity.

9.2 Exact classification inside the affine family

Theorem 9.4 (Affine-family main classes). *For $p \geq 5$, the loops $Q_p(a), Q_p(a')$ are main-class-equivalent exactly when*

$$a' \in \mathcal{A}(a) = \{a, 1 - a, 1/a, 1/(1 - a), a/(a - 1), (a - 1)/a\}.$$

Distinct parameters are never isomorphic; each loop has automorphism group $\text{AGL}(1, p)$.

Proof. Swapping the inputs and solving the relation for either input give the displayed parameters, including all triples involving ∞ . Thus these relations suffice.

The types of row pairs involving infinity and of two finite rows are

$$(2, \text{ord}_p(b)^{(p-1)/\text{ord}_p(b)}), \quad ([a^{-1}]_p, p + 1 - [a^{-1}]_p).$$

They coincide only if $\text{ord}_p(b) = p - 1$ and $[a^{-1}]_p \in \{2, p - 1\}$. The column version exchanges a, b . Outside $E = \{2, 1/2, -1\}$ both identity lines are intrinsic: each is the unique line with all p incident pair types the same, whereas a finite line has both types. An isotopy between such loops must fix the two identity lines; the loop equations make its three maps equal. Its finite-field bijection f satisfies

$$f(ax + by) = a'f(x) + (1 - a')f(y),$$

also at $x = y$. Subtract $f(0)$ and set one argument to zero. Inverting a, b in the resulting identities proves additivity of $h(x) = f(x) - f(0)$. On a prime field h is multiplication by a nonzero scalar; the identity then forces $a = a'$.

For a paratopy first perform its coordinate permutation, replacing a by $t \in \mathcal{A}(a)$. The set E is a single orbit. If exactly one of t, a' is in E , either its special and finite pair types coincide in some view, giving an invariant mismatch, or both identity lines remain intrinsic and the same argument forces equality. If both are in E , they already lie in the same displayed orbit. This proves necessity. For an isomorphism the identity is fixed without exceptions, so the additive argument always gives $a = a'$ and $f(x) = ux + v$. Conversely every such map with $u \neq 0$, fixing ∞ , is an automorphism. $\square$

Let S_p be the full parameter set satisfying (7). The anharmonic orbits have size six, apart from E of size three and the two roots of $a^2 - a + 1 = 0$ when present. Let e_p, h_p indicate whether these exceptional orbits lie in S_p . The exact number of represented FFF main classes is

$$C_p^{\text{aff}} = (|S_p| + 3e_p + 4h_p)/6.$$

Indeed, the three involutions fix only $1/2, 2, -1$, and the order-three elements fix exactly the two polynomial roots. For $p \geq 5$ these exceptional sets are disjoint. Consequently, Theorem 9.3 gives more than $p/6000$ FFF main classes at every prime-successor order $p + 1$ with $p \geq 10^{12}$.

Finite controls construct all 253 parameters for the 13 odd primes through 43, compare all views with the criterion, and check parameter transformations and orbit counts. They do not prove the analytic estimate or its uniform cutoff. At $p = 17$ the criterion excludes

only the scalar affine family, not all order-18 squares. The classification is within this family, not a census of all Latin squares. Literature priority and external expert review of the analytic application remain open.

10 Row-fibred extensions and represented main classes

The following construction allows the inner operation to depend on an outer row. Related index-two constructions already occur in the loop literature, for example Johnson's dihedral extension [Joh06]. No priority claim for the construction, its FFF consequences, or the bounds below is made here.

10.1 An exact three-view pattern formula

Let $B : Y \times Y \rightarrow Y$ be Latin and, for each $y \in Y$, let $Q_y : X \times X \rightarrow X$ be Latin. Define

$$M((r, y), (c, z)) = (Q_y(r, c), B(y, z)). \quad (9)$$

All sets are finite and nonempty. The fibre squares need not be isotopic.

Theorem 10.1 (Row-fibred pattern formula). *The table M is Latin, and in the three named views*

$$\text{pat}(M) = \text{pat}(B) \text{ OR } \bigvee_{y \in Y} \text{pat}(Q_y).$$

In particular, M is FFF if and only if B and every Q_y are FFF.

Proof. In a fixed row, both coordinate maps in (9) are bijections. In a fixed column, the second output determines y and then the first determines r , proving the Latin property.

For rows $(r, y), (r', y')$ with $y = y'$, the induced permutation fixes the outer column z and restricts on each such layer to the corresponding row-pair permutation of Q_y . With $y \neq y'$, it projects to the row-pair permutation of B on outer columns. For columns $(c, z), (c', z')$ with $z = z'$, the analogous permutation fixes y and restricts to a column-pair permutation of Q_y ; with $z \neq z'$ it projects to the column-pair permutation of B . For symbols $(s, t), (s', t')$ with $t = t'$, the outer column z is fixed, and the outer row is the unique y such that $B(y, z) = t$. The induced map on inner columns is the symbol-pair permutation of Q_y . Every y occurs for one z . With $t \neq t'$, the outer columns instead follow the symbol-pair permutation of B .

Every fibre witness therefore embeds in the same view of M . In every cross-layer case an upstairs cycle has length divisible by its projected cycle length. Thus, if both the quotient and all fibres are F in that view, no odd cycle is possible upstairs. To finish the equality, a quotient odd cycle must be lifted without increasing its length by an even factor. This requires a choice of inner coordinates.

For quotient rows y, y' , fix an inner column c and an inner row r . Choose r' uniquely so that $Q_y(r, c) = Q_{y'}(r', c)$. The induced upstairs row-pair map fixes c , so every quotient cycle lifts with its original length. For quotient columns z, z' , choose upstairs columns with a common inner column c and fix an inner symbol s . At each outer row y on the cycle, choose r_y by $Q_y(r_y, c) = s$. The induced map carries (r_y, y) around precisely the quotient cycle. Finally, for quotient symbols t, t' , take a common inner symbol s and an inner column c . The symbol-pair map keeps c fixed and follows the quotient symbol cycle on z : in the relevant row y , the inner row is determined by $Q_y(r, c) = s$. Hence the quotient odd cycle survives in this view as well. $\square$

For $|X|, |Y| > 1$, the construction cannot produce FFF of order 18: one factor of every such factorization of 18 is odd and greater than one. The order-one square is vacuously FFF and is not an odd-order obstruction.

10.2 Cycle spectra and the order-20 input

For an unordered pair of distinct lines in a view, sort its cycle lengths as a partition λ . Let $\Sigma_L(\lambda)$ count all such pairs, summed over the three views. Isotopy conjugates or inverts pair permutations; parastrophy also permutes the views. Thus Σ_L is a main-class invariant. Sorting each partition before comparison is essential.

Proposition 10.2 (Product spectra and intercalates). *Let $I(L)$ denote the number of intercalates in a Latin square of order n , and let K have order m . Then*

$$I(L \times K) = m^2 I(L) + n^2 I(K) + 4I(L)I(K).$$

If L is FFF, and D duplicates each part of a partition, then

$$\Sigma_{L \times C_2} = 4D_*\Sigma_L + 3n[2^n]. \quad (10)$$

In particular, distinct spectra remain distinct after any number of products with C_2 .

Proof. A two-cycle in a row-pair permutation is one intercalate. If product rows differ only in the L coordinate, the common K row has m choices and each two-cycle has m copies, giving $m^2 I(L)$. The other one-coordinate case gives $n^2 I(K)$. If both coordinates differ, a pair of factor two-cycles has two alignments of the product rows, each producing two two-cycles, giving the last term. Distinct Latin lines have no fixed points, so there are no other ways to produce a two-cycle.

For (10), in each view the n product-line pairs with equal L coordinate have type 2^n . Each unordered distinct L line pair has four product-line pairs. A cycle of length a times a cycle of length b has $\gcd(a, b)$ cycles of length $\text{lcm}(a, b)$. Since every L part is even, both multiplication by the identity on two points and by its transposition duplicate every part. Summing the three views proves the formula. The map D is injective on partitions, so its pushforward is injective. $\square$

Computational Theorem 10.3 (Frozen order-20 family). *One specified cycle-union trade family of an FFF Steiner-loop table of order 20 represents exactly 512 main classes within that family. Two separately specified quadratic tables represent further classes. Consequently at least 514 FFF main classes of order 20 exist.*

The finite evidence enumerates all $2^{10} = 1024$ masks of the ten transpositions in one row pair. Every resulting table is Latin and FFF under two independent physical three-view scanners. Exactly 512 sorted spectra occur, each at complementary masks. Complementing a mask swaps the two rows globally, so it supplies the upper bound of 512 classes as well as the invariant's lower bound. The two quadratic controls have distinct spectra outside that family; their intercalate counts are 190 and 19, whereas the trade-family counts range from 542 to 676. These are exact table comparisons, not a census of all order-20 squares. Equation (10) alone propagates the lower bound 514 to orders $20 \cdot 2^d$.

10.3 Recovering quotient layers

A *binary three-sorted quotient* of Q is a triple of surjective maps a, b, d from its row, column and symbol sets to $\mathbb{F}_2$ satisfying $d(Q(r, c)) = a(r) + b(c)$. The independent labelling of the three sorts is part of the definition. The corresponding partitions, not their chosen bit labels, are intrinsic.

Lemma 10.4 (Binary quotient descent). *Suppose no Q_y in (9) has a binary three-sorted quotient. Every such quotient of M is constant on each inner layer and descends to a binary quotient of B ; conversely, each binary quotient of B pulls back to one of M .*

Proof. First, any three binary maps satisfying $d(Q(r, c)) = a(r) + b(c)$ are either all constant or all surjective. If a is constant, fixing c and varying r shows that d is constant, and then b is constant. The argument with a, b exchanged is the same; if d is constant, varying either input shows both others constant.

Restrict a quotient triple of M to the block of rows at y , columns at z , and symbols at $B(y, z)$. It satisfies this equation on Q_y . Quotient freeness makes all three restrictions constant. Running over y, z makes the full maps constant on every inner layer, and their constants satisfy the quotient equation on B . Surjectivity is preserved by descending or pulling back. $\square$

For $B = E = (\mathbb{F}_2)^d$, every binary quotient has the form

$$a(y) = \chi(y) + a(0), \quad b(z) = \chi(z) + b(0), \quad d(t) = \chi(t) + a(0) + b(0),$$

where $\chi : E \rightarrow \mathbb{F}_2$ is a nonzero linear character. This follows by evaluating the quotient equation at $(y, 0)$ and $(0, z)$ and subtracting the constants. All $2^d - 1$ characters separate points of E . Their common refinement therefore recovers each inner layer intrinsically, in all three sorts.

Theorem 10.5 (Affine-orbit classification of a fixed family). *Fix representatives $Q_1, \dots, Q_m$ of pairwise distinct main classes of order n , none admitting a binary three-sorted quotient. For each function $f : E \rightarrow \{1, \dots, m\}$ define*

$$M_f((r, y), (c, z)) = (Q_{f(y)}(r, c), y + z).$$

Two members M_f, M_g are main-class-equivalent if and only if $g(Ay + u) = f(y)$ for some $A \in \text{GL}(d, 2)$ and $u \in E$.

Proof. By Lemma 10.4, every main-class equivalence preserves the intrinsic layers, up to a permutation of their three sorts, and induces a paratopism of the outer table of E . Its blocks are colored by the main classes $[Q_{f(y)}]$. A nonconstant coloring is constant along an outer row but not along an outer column or symbol line. Thus a paratopism between nonconstant members must carry the source row sort to the target row sort. A constant coloring can match only the same constant coloring, since the input main classes are distinct.

Every paratopism of $y + z = t$ acts affinely on each sort with a common invertible linear part. Indeed, permuting its three coordinates preserves this relation in characteristic two. After undoing that permutation, write $\gamma(y + z) = \alpha(y) + \beta(z)$. Setting $u = \alpha(0)$, $v = \beta(0)$ gives $\alpha(y) = Ay + u$, $\beta(z) = Az + v$ and $\gamma(t) = At + u + v$, with A additive and bijective. The transported block colors now give $g(Ay + u) = f(y)$.

Conversely, this equality makes the three maps

$$(r, y) \mapsto (r, Ay + u), \quad (c, z) \mapsto (c, Az + v), \quad (s, t) \mapsto (s, At + u + v)$$

an explicit isotopism for any $v \in E$. The fixed choice of the representatives is used in this sufficiency argument. $\square$

Corollary 10.6 (Constructed class counts). *Let $c(h)$ be the number of cycles of h on E . The fixed family in Theorem 10.5 represents exactly*

$$T_d(m) = \frac{1}{|\text{AGL}(d, 2)|} \sum_{h \in \text{AGL}(d, 2)} m^{c(h)}, \quad |\text{AGL}(d, 2)| = 2^d \prod_{i=0}^{d-1} (2^d - 2^i) \quad (11)$$

main classes. If the inputs are FFF, these are FFF classes and hence give a lower bound for the total at order $n2^d$. A weaker bound is $\binom{m+2^d-1}{2^d}$, obtained by distinguishing input multisets.

Proof. Burnside’s lemma counts colorings fixed by each affine permutation: they are precisely the colorings constant on its cycles, numbering $m^{c(h)}$. Affine orbits refine the input multisets. The FFF assertion follows from Theorem 10.1. $\square$

Computational Theorem 10.7 (Specialization from order 20). *Using the computational exclusion of order 10 in the full companion report (register entry C157), the 514 inputs in Computational Theorem 10.3 are binary-quotient-free. Therefore there are at least $T_d(514)$ FFF main classes of order $20 \cdot 2^d$. The first four values are:*

Order	Proven lower bound, not a census
40	132355
80	2942384005
160	3625556867870760586
320	73586711420426563009097417622853741216

A binary quotient of an order-20 square has balanced fibres: the Latin bijections identify the two color-class sizes, so they are 10 in each sort. Each block is a Latin subsquare of order 10. Subsquare monotonicity would make it FFF, contrary to C157. Equation (11) now proves the conditional construction count; an exact enumeration of the affine permutation groups through dimension four supplies the displayed integers. No physical enumeration of that many output squares is asserted.

Two checked order-160 outputs even have the same input multiset and the same sorted cycle spectrum, but their four-point color sets are an affine plane and an affine tetrahedron. Their affine spans have different dimensions, so Theorem 10.5 separates their main classes. This demonstrates a limitation of the spectrum invariant, not its completeness.

The count is exact only within the fixed family. It does not classify all higher-order FFF squares, justify arbitrary direct-product cancellation, or identify all extensions obtained by changing the chosen representatives within their main classes. The binary-quotient-free hypothesis must be checked on the inputs; it cannot be iterated blindly on outputs that already have binary quotients.

11 An exact position–value coloring formulation

This section gives an equivalent finite constraint model, not a decision of any new order. In one view let each line a be a bijection $f_a : P \rightarrow Q$. For distinct a, b the permutation $\pi = f_b^{-1} f_a$ is fixed-point-free. A permutation has only even cycles if and only if its points admit a binary coloring that toggles along every edge: alternating a coloring around a cycle is consistent exactly when its length is even.

Theorem 11.1 (Position–value channel). *Introduce colors $e_i, h_v \in \{0, 1\}$ for positions $i \in P$ and values $v \in Q$. The equations*

$$e_i = h_{f_a(i)}, \quad e_i = 1 - h_{f_b(i)} \quad (i \in P) \quad (12)$$

are satisfiable if and only if π has only even cycles. Every solution has exactly $n/2$ ones in each color family. One may add $e_{i_0} = 0$ at any chosen position without changing satisfiability.

Proof. For $j = \pi(i)$, equality $f_a(i) = f_b(j)$ makes (12) imply $e_j = 1 - e_i$. Conversely, from a toggling coloring set $h_v = e_{f_a^{-1}(v)}$. The first equation is immediate. The second follows because $f_a^{-1}f_b = \pi^{-1}$ and toggling also holds on inverse edges. Since the lines are bijections, summing the two equations gives $\sum_i e_i = \sum_v h_v = n - \sum_v h_v = n/2$. Complementing all e and h simultaneously preserves the equations and permits the single gauge condition. $\square$

For a Latin cell variable X_{rcs} asserting $L(r, c) = s$, the antecedents A, B for $f_a(i) = v$ and $f_b(i) = v$ are:

View	Position	Value	A	B
row	column i	symbol v	X_{aiv}	X_{biv}
column	row i	symbol v	X_{iav}	X_{ibv}
symbol	column i	row v	X_{via}	X_{vib}

Writing $e = e_i$ and $h = h_v$, four clauses implement the equations:

$$(\neg A \vee \neg e \vee h), \quad (\neg A \vee e \vee \neg h), \quad (\neg B \vee e \vee h), \quad (\neg B \vee \neg e \vee \neg h).$$

With Latin exactly-one constraints, imposing these clauses independently for every line pair in every view is equivalent to FFF. The reverse implication extends each FFF table by its pairwise alternating colorings; no coupling of arbitrary gauges is required. Isotopy invariance permits reduced tables in an existence search.

Each line pair uses $2n$ colors, $4n^2$ conditional clauses, and one optional gauge unit. Balance follows from the equations and requires no cardinality auxiliaries. Frozen positive controls at orders 8, 12, and 14 and checked negative order-6 controls validate the implementation. In one fixed order-18 row context, the recorded model has 22356 variables and 745065 clauses, compared with 270216 and 3190617 in the prior physical-arc representation. Both bounded 60-second solver probes timed out. This is a size reduction, not evidence of faster solution or of nonexistence at order 18.

12 Computational reproducibility

The research repository separates proofs, frozen data, programs, commands, logs, and SHA-256 manifests. A companion reproducibility quickstart supplies the complete command-to-output map. The compact paper uses the following packages.

This edition’s additional evidence

This is a dated local working-copy snapshot, not a claim that the public repository or a persistent archive already contains it. Its source directory `manuscript/candidates/2026-09-27_research_snapshot/` contains a theorem-to-evidence map, the captured proof notes and small comparison objects, the build commands, and a dedicated audit. The

capture lists source hashes and the claim cutoff C251. All inherited sources and earlier release PDFs are preserved.

The principal additions are the affine-family criterion and classification (C165–C167), spectrum and row-fibred formulas (C246–C247), binary quotient recovery and affine-orbit counting (C248–C250), and the exact position–value channel (C245). The order-20 source audit checks the complete 1024-mask family, not a sample. The higher-order class bounds are rigorous consequences of that source evidence and C157, not physical scans of all the outputs being counted. The two order-160 controls are compared by exact sorted spectra, not only by hash equality.

The new finite audit records which controls were freshly rerun and which earlier certified computations were only verified against frozen reports. In particular, the expensive C157 master searches and the scoped order-18 solver proofs are not silently reported as fresh reruns of this paper build. Historical sign and manuscript audit tools certify their own dated scope, not every new theorem automatically.

SHA-256 provides byte identity, not correctness of mathematics, source completeness, peer review, or literature priority. The source archive’s checksum manifest excludes itself and uses a separate output manifest for the generated PDFs and reports. The new PDFs are local review outputs; the two curated historical PDF exceptions in Git are not replaced by this task.

Trade parity `repro_runs/2026-08-10_paper_symbol_sign_audit/`. The exact control checks 1028307 row-, column-, and symbol-view trades on all reduced Latin squares of orders 2, 4, and 6, with zero failures. The proof, not this computation, establishes the general theorem.

Small orders `repro_runs/2026-04-26_small_order_witness_reruns/`. The exact-cover generator freshly enumerates 1, 4, and 9408 reduced squares of orders 2, 4, and 6. Every table is checked Latin, reduced, and unique. The generated order-6 set agrees exactly with the official ANU file.

Order-8 census `repro_runs/2026-04-26_order8_core/`. The frozen 5140393-byte input has SHA-256 `e5980ff5bc4ac46cbbefd1f1333258d0905724c42a976c4afe61ea501a4ccf87`. The theorem-relevant output `results/order8_recheck_fullscan_results.json` has SHA-256 `10f2f43c8dcaa8ec93f83abf6b768efa47f909a9c8623d4bc648e6da097ccd39`.

Group-isotopy closure `repro_runs/2026-04-26_order8_closure_reconstruction/`. This independently reconstructs the basis-family closure scan and reproduces the 5/225 split.

Direct products `repro_runs/2026-04-26_direct_product_fff/`. The validator checks the product-permutation formula in all three views and tests the group-isotopy equivalence on group and non-group examples.

Explicit non-power-of-two witnesses `repro_runs/2026-09-23_fff_nonpower_witness_intake/`. An independent standard-library scanner checks the frozen order-12, 14, and 36 tables as Latin, traverses every induced cycle in all three views, and records exact hashes. The displayed reduced order-12 table is `data/order12_first.json`, SHA-256 `8b71d6ecaa6ab86a806a0660529b404d8f3667db2b58b82bf5c72fed010fa416`.

Three-point lift controls `repro_runs/2026-09-23_fff_three_point_lift_audit/`.

The script checks subgroup-coset moments for five odd parameters and scans three views at orders 56, 80, and 240. The universal bound is proved in Theorem 7.3; finite controls do not replace it.

Subsquare and congruence obstructions `repro_runs/2026-08-09_fff_subsquare_congruence_obstructions/`. The package scans all subsquares of the 230 frozen FFF representatives, exhausts the binary twists used as controls, and validates the explicit twisted order-8 loop.

Binary line kernel and half-order subsquares `repro_runs/2026-08-11_fff_binary_homotopy_subsquares/`. The package compares the subsquares constructed from every nonuniversal binary line dependency with an independent direct enumeration on 9778 tables. All predicted and observed sets agree exactly, and the semantic verifier reproduces the frozen order-8 distribution with zero errors.

Prime-field line quotients `repro_runs/2026-08-11_fff_prime_field_line_quotients/`. The audit checks five prime fields on 9778 tables, verifies all 1160 nonuniversal quotient classes, and independently reconstructs 48890 modular ranks with zero errors. The theorem itself is proved separately.

External-input release audit `repro_runs/2026-08-09_paper_release_audit/`. It downloads the cited ANU and Wanless inputs afresh and verifies exact byte matches to the frozen files. As with any computational theorem based on an external census, completeness of the published enumeration remains a cited scientific dependency.

Every package contains a README, exact commands, environment record, input and output manifests, and a package-level SHA-256 manifest. The command

```
python3 repro_runs/2026-08-10_paper_symbol_sign_audit/scripts/audit_manuscript_computational_claims.py
```

cross-checks the historical September 16 long manuscript. The compact draft retains its relevant computational statements and adds explicit witness and constructed-class statements. These require the additional dated evidence and checks described above.

Generated PDFs, solver binaries, virtual environments, and raw chat exports are not part of the tracked source archive. No timeout, partial solver model, or partial order-10 diagnostic is used as proof evidence in this compact paper. The complete C157 master computation is a different, explicitly imported dependency.

13 Discussion and open problems

The former assertion that FFF squares of order $n > 1$ exist exactly at powers of two is false: the displayed order-12 witness is a counterexample. Nontrivial odd orders, order 6, and order 10 are excluded; the order-10 argument is an explicit computational dependency imported from the full companion report. Order 18 remains the next undecided small even non-power-of-two order.

The row-fibred theorem preserves the exact three-view pattern even with different fibres. Under the additional binary-quotient-free hypothesis, Theorem 10.5 classifies the fixed elementary-abelian-quotient family by affine orbits. Computational Theorem 10.7 gives lower bounds for all FFF classes, not exact censuses or unrestricted direct-product cancellation.

The next mathematical questions are:

1. Decide unrestricted FFF existence at order 18 and sharpen the minimal dyadic-exponent function $h(m)$.
2. Find structural obstructions coupling all three views that do not also exclude the explicit order-12, 14, and 98 controls.
3. Determine what lies outside the fixed constructive families and whether quotient recovery survives weaker hypotheses.

The full report records several exact, but restricted, order-18 construction exclusions. None covers all order-18 Latin squares. Timeouts in the smaller equivalent coloring model are likewise not exclusions. No global Alon–Tarsi conclusion is claimed [AT92]; literature priority and independent expert review remain open.

References

- [AT92] Noga Alon and Michael Tarsi. Colorings and orientations of graphs. *Combinatorica*, 12(2):125–134, 1992. <https://doi.org/10.1007/BF01204715>.
- [Bér26] Gergely Bérczi. Evolving local corrections for global constructions in combinatorics, 2026. arXiv:2603.06692, <https://arxiv.org/abs/2603.06692>.
- [CFG⁺10] Charles J. Colbourn, Anthony D. Forbes, Mike J. Grannell, Terry S. Griggs, Petteri Kaski, Patric R. J. Östergård, David A. Pike, and Olli Pottonen. Properties of the steiner triple systems of order 19. *The Electronic Journal of Combinatorics*, 17(1):R98, 2010. Primary STS(19) classification source; the local order-20 FFF witness is checked directly without reproducing its census.
- [CS26] Yangyang Cheng and Amedeo Sgueglia. The perfect 1-factorisation conjecture holds asymptotically, 2026. Introduction recalls the classical polygon-and-centre factorisation for prime order; not a source for the three-view FFF criterion.
- [EG24] Grahame Erskine and Terry S. Griggs. Cycle switching in steiner triple systems of order 19, 2024. Section 3.2 prints the 11 rotational starters of the no-6-cycle STS(19).
- [Joh06] K. W. Johnson. The construction of loops using right division and ward quasigroups. *Quasigroups and Related Systems*, 14(1):27–41, 2006. Related extension constructions; not an attribution of the FFF results.
- [Kat08] Nicholas M. Katz. Estimates for mixed character sums. *Geometric and Functional Analysis*, 18(4):1251–1269, 2008. Theorem 1.5; primary text checked 27 September 2026.
- [McK] Brendan D. McKay. Latin squares: Combinatorial data. Australian National University data page. <https://users.cecs.anu.edu.au/~bdm/data/latin.html>, accessed 2026-08-09.
- [MMM07] Brendan D. McKay, Alison Meynert, and Wendy Myrvold. Small latin squares, quasigroups, and loops. *Journal of Combinatorial Designs*, 15(2):98–119, 2007. <https://doi.org/10.1002/jcd.20105>.
- [Moo91] G. Eric Moorhouse. Bruck nets, codes, and characters of loops. *Designs, Codes and Cryptography*, 1(1):7–29, 1991. Theorem 4.2. Author-hosted article copy: <https://ericmoorhouse.org/pub/loop1.pdf>; separate publications/errata listing: <https://ericmoorhouse.org/pub/>; accessed 2026-08-27. DOI: <https://doi.org/10.1007/BF00123956>.
- [Wan04] Ian M. Wanless. Cycle switches in latin squares. *Graphs and Combinatorics*, 20:545–570, 2004. Proposition 1 gives the row-, column-, and symbol-parity changes under cycle switching; <https://doi.org/10.1007/s00373-004-0567-7>.
- [WI05] Ian M. Wanless and Edwin C. Ihrig. Symmetries that latin squares inherit from 1-factorizations. *Journal of Combinatorial Designs*, 13:157–172, 2005. Section 3 relates two-line cycle lengths to pairwise unions of 1-factors; <https://doi.org/10.1002/jcd.20045>.